GLENCORE

Maintenance Work Management Guideline

1. WORK IDENTIFIED, REGISTERED & PRIORITISED

08

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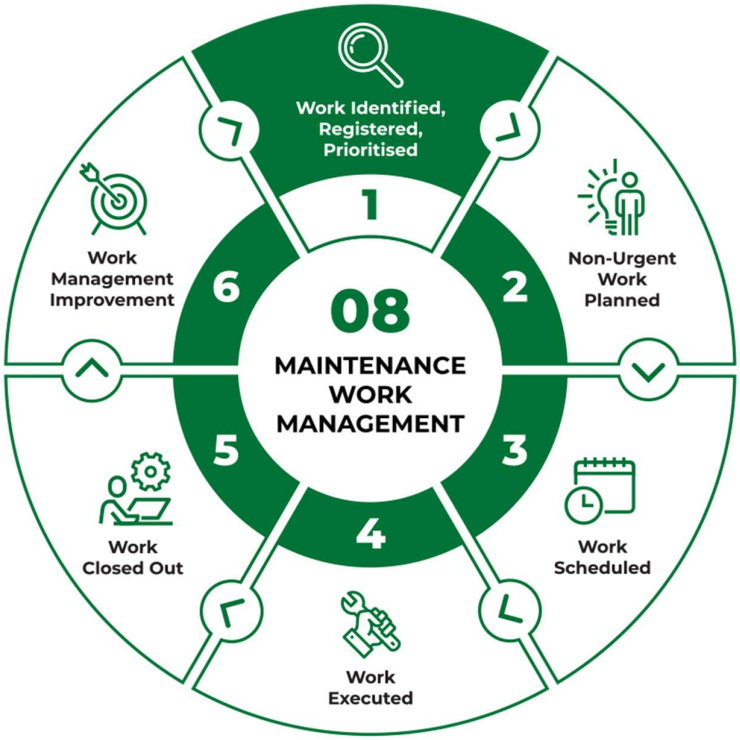
Sub-Process Overview

Purpose

The Glencore Copper Asset Management Framework (AMF) enables the business to deliver its Asset Management purpose and vision, as defined in the AMF Policy. The Framework consists of 18 Elements, detailed in Manuals that define how Glencore Copper manages the whole life of its assets to optimise value and ensure assets are capable of meeting the business and stakeholder objectives.

Maintenance Work Management is Element 08 of the AMF and defines effective work management practices and performance requirements to enable work to be performed safely, efficiently, with precision and when required.

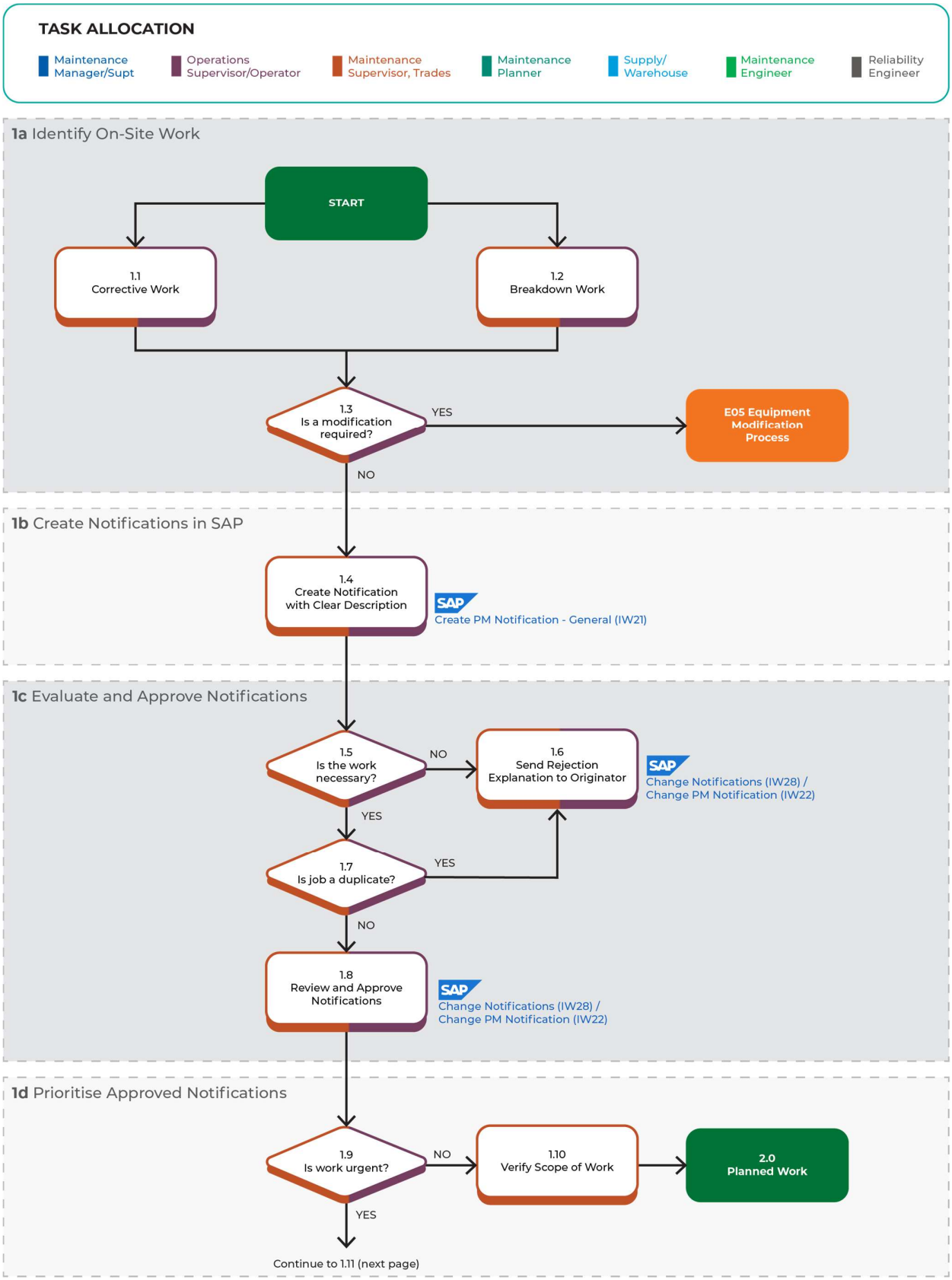
This document is intended to provide end users with understanding and knowledge of sub-process **1. Work Identified, Registered & Prioritised**, including how corrective actions are identified, prioritised, and recorded in SAP.

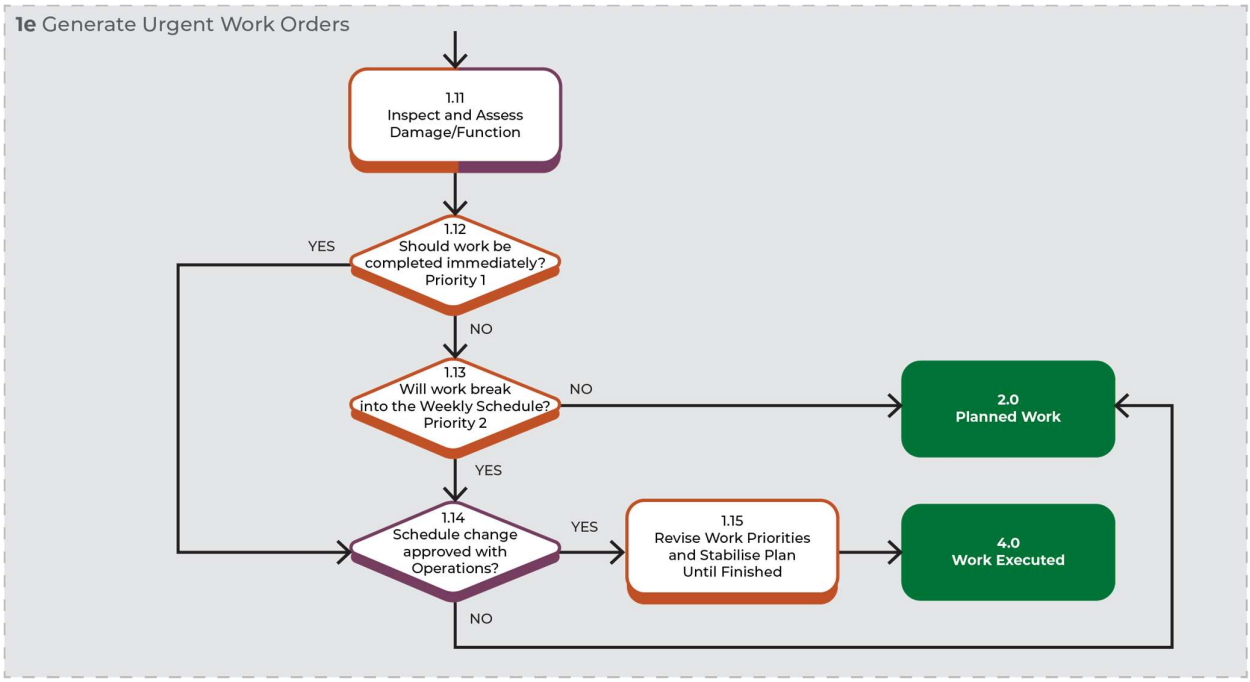


Objectives & Flowchart

The objective of sub-process **1. Work Identified, Registered & Prioritised**, is to ensure that all potential maintenance tasks are identified early, well described and correctly prioritised. This will ensure that:

- Work duplication is avoided
- Correct work priorities are executed
- Minimise break-ins to the weekly schedule
- Maximise the efficiency of subsequent planning and scheduling steps


TASK ALLOCATION
Maintenance Manager/Supt
Operations Supervisor/Operator
Maintenance Supervisor, Trades
Maintenance Planner
Supply/ Warehouse
Maintenance Engineer
Reliability Engineer
1a Identify On-Site Work
START
1.1 Corrective Work
1.2 Breakdown Work
1.3 Is a modification required?
YES
E05 Equipment Modification Process
NO
1b Create Notifications in SAP
1.4 Create Notification with Clear Description
SAP
Create PM Notification - General (IW21)
1c Evaluate and Approve Notifications
1.5 Is the work necessary?
NO
1.6 Send Rejection Explanation to Originator
SAP
Change Notifications (IW28) / Change PM Notification (IW22)
YES
1.7 Is job a duplicate?
YES
NO
1.8 Review and Approve Notifications
SAP
Change Notifications (IW28) / Change PM Notification (IW22)
1d Prioritise Approved Notifications
1.9 Is work urgent?
NO
1.10 Verify Scope of Work
2.0 Planned Work
YES
Continue to 1.11 (next page)


1e Generate Urgent Work Orders
1.11
Inspect and Assess
Damage/Function
1.12
Should work be
completed immediately?
Priority 1
YES
NO
1.13
Will work break
into the Weekly Schedule?
Priority 2
NO
2.0
Planned Work
YES
1.14
Schedule change
approved with
Operations?
YES
1.15
Revise Work Priorities
and Stabilise Plan
Until Finished
4.0
Work Executed
NO

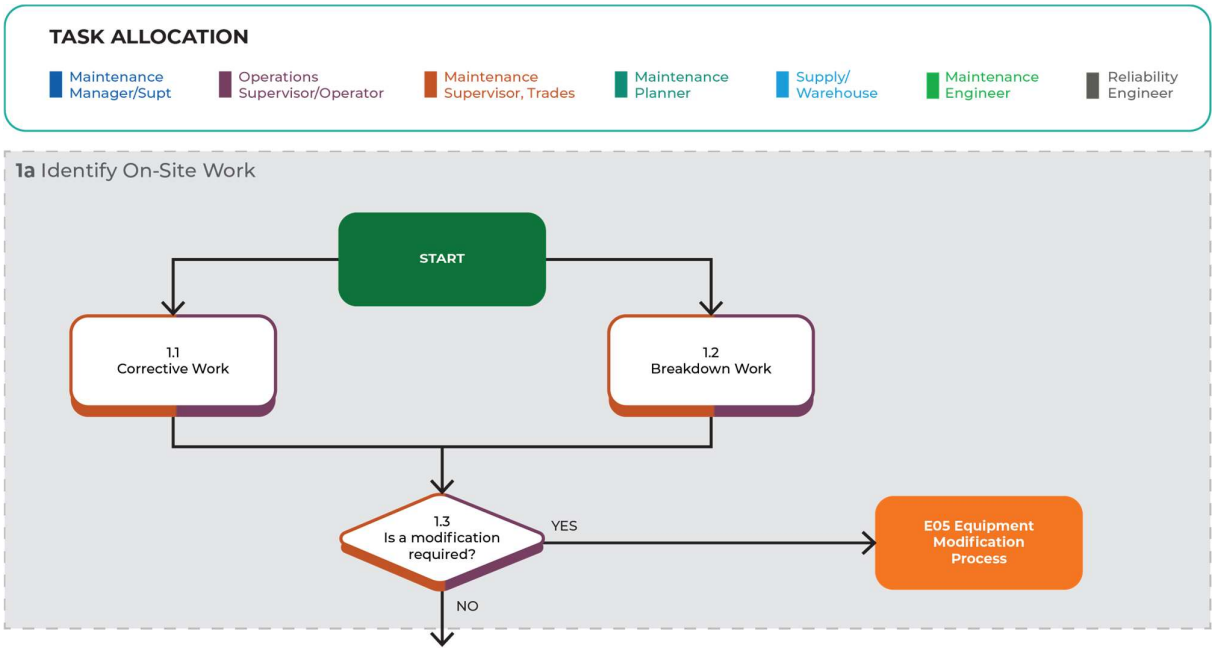
Roles & Responsibilities

R Responsible
S Support
I Inform

		Operations			Maintenance				Other		
		Superintendent	Supervisor	Operator	Superintendent	Supervisor	Trades	Planner	Maintenance Engineer	Reliability Engineer	Supply / Warehouse
1a	**Identify on-site work** (1.1 & 1.2)		R			R		R			
1.3	Is a modification required?		R			R		R			
1.4	Create Notification with clear description		R			R		R			
1.5	Is the work necessary?		R	S		R	S				
1.7	Is job a duplicate?		R			R					
1.6	Send rejection explanation to Originator		R	I		R	I				
1.8	Review & approve notifications		R			R					
1.9	Is work urgent?					R	S				
1.10	Verify Scope of Work					R	S				
1.11	Inspect and Assess Damage / Function					R	S				
1.12	Should work be completed immediately? (Priority 1)		R	S		S					
1.13	Will work break into the weekly schedule? (Priority 2)					R		I			
1.14	Schedule change approved with Operations?		R	S		S					
1.15	Review work priorities and stabilise plan until finished					R					

1. Work Identified, Registered & Prioritised

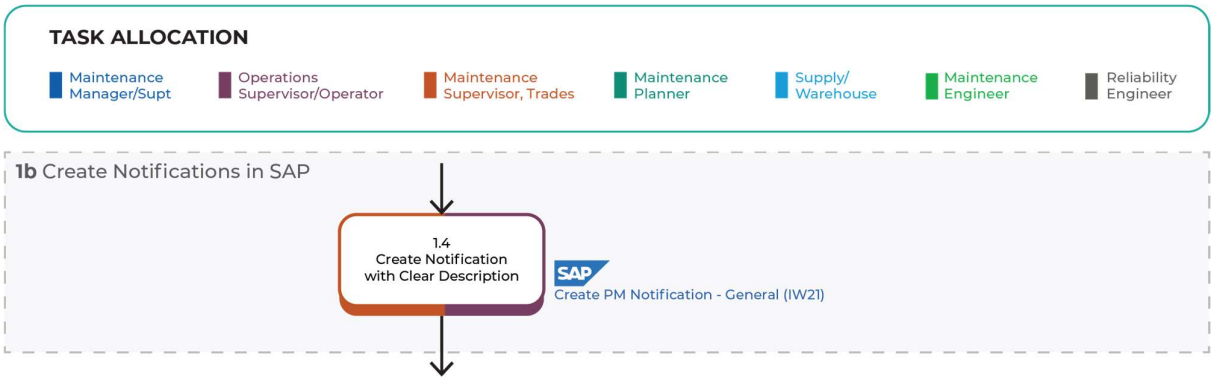
1a Identify On-Site Work



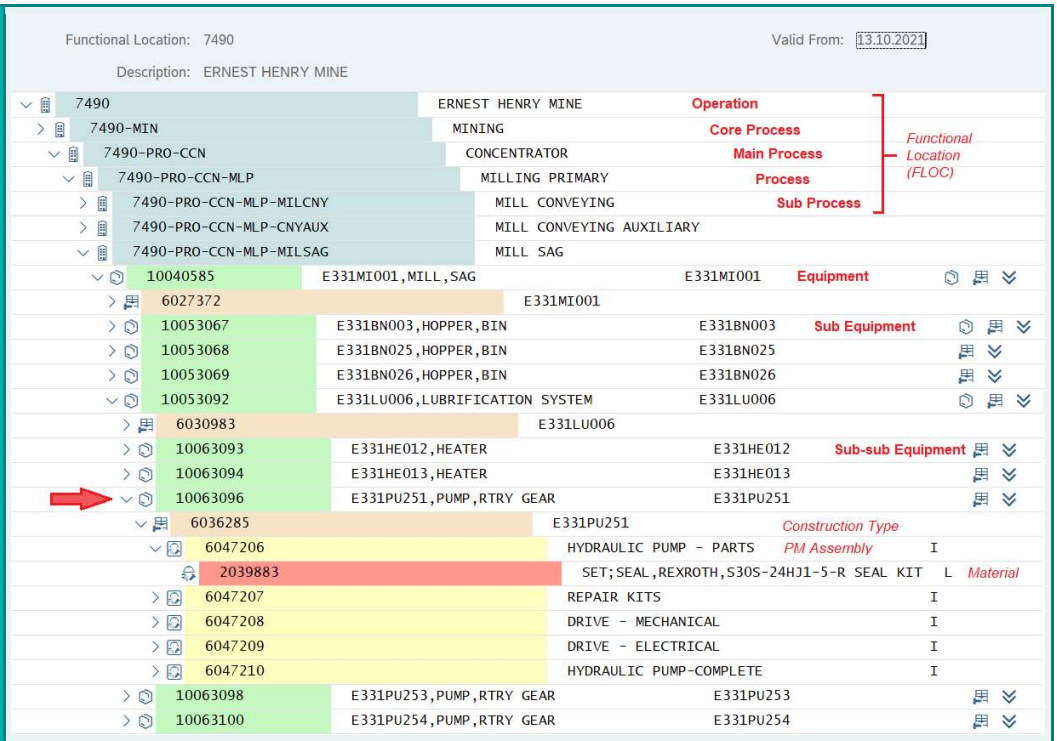
What?	Identify work to be completed.
Why?	Early and effective identification of work is a key component of preventative maintenance cultures.
Who?	• Tradespersons & Supervisors • Operators & Supervisors • Reliability Engineers & Technicians • Maintenance Planners • HSE representatives • External Service Providers and Consultant engagements • Regulator (i.e. Department of Resources)
How?	Work can be identified through the following activities: • Operations inspections & equipment pre-starts • Routine preventive maintenance (services & inspections) • Execution of corrective maintenance • Condition monitoring (predictive) • Statutory inspections & reports • HSE inspections, incident investigations & reports If modification of equipment is required (i.e., a significant change from the current design), proceed to the **Equipment Modification Process** (change management) in **Element 05 – Equipment Operating & Maintenance Strategy** of the AMF.

	If no equipment modification is required, proceed to next step **1b Create Notifications in SAP.**
When?	Daily, through work routines of all stakeholders.

1b Create Notifications in SAP



What?	Enter the work required into SAP by creating a Notification.
Why?	By capturing all work in SAP, the prioritising, planning and scheduling of work can be optimised. New Notifications will be extracted and reviewed daily by frontline Supervisors. Information in the Notification is also critical for future reliability studies and defect elimination work.
Who?	• Tradespersons & Supervisors • Operators & Supervisors • Reliability Engineers & Technicians • Maintenance Planners • HSE representatives • General users
How?	Create a Notification in SAP using **Create PM Notification – General (IW21)**. Refer to **SAP PM QRG – Create PM Notification** for further details on using this transaction. • Select the correct type of Notification to raise. Almost all manually created Notifications should be M1 type (Maintenance Request). Refer to **SAP PM QRG – Create PM Notification** for more detail on Notification types. • Notifications should always be raised at the lowest level possible (lowest maintainable item or function) in the Functional Location Structure. This typically would be the "Sub-Equipment" level or lower, where these levels exist and it is appropriate. ○ Following is a typical example showing the structure for a SAG Mill. In this example, if the identified problem was with hydraulic pump PU251 in the lubrication system, you would raise the notification against the Sub-sub Equipment as indicated by the red arrow.



- Confirm that the notification is not a duplicate by checking the last three notifications created against the selected equipment, as detailed in **SAP PM QRG – Create PM Notification**. Do not raise a new notification where one already exists for the same job.

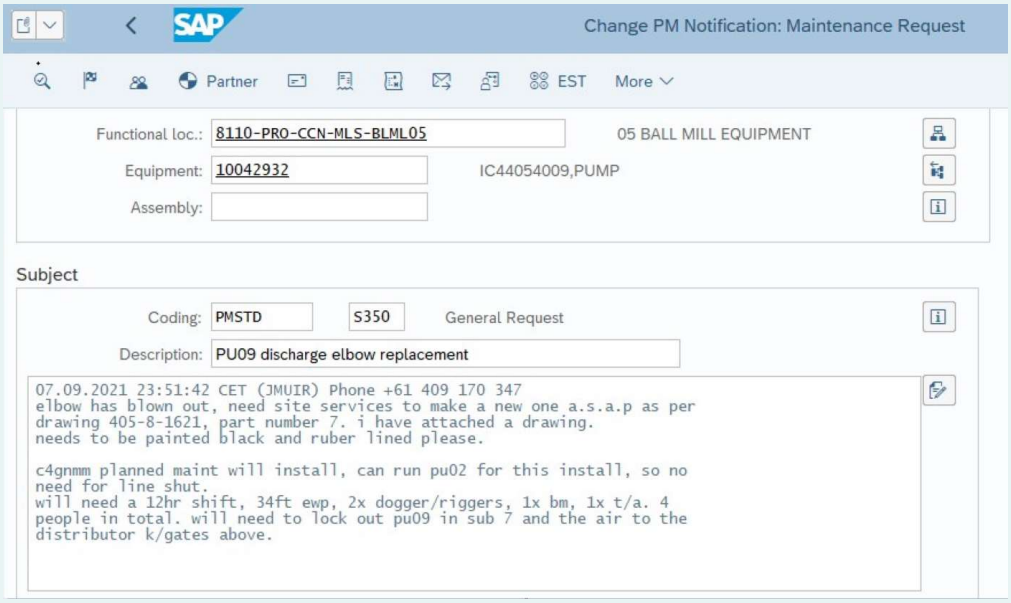
As much information as practical should be entered into the Notification describing the following:

- Short text description of what work is required.
- Long text description including:
  - Detailed description of the problem and system status, including root cause if known.
  - Scope of work required / proposed solution. This should include any special pre or post activities to allow completion of the work.
  - Identification of materials (parts) and any consumables required to complete the work (refer **SAP PM QRG – Material Search**).
  - Estimate of what trade or external supplier skills are required and the hours needed.
  - List any other resources required (e.g. scaffold, crane, hydroblasting etc).
  - List of any specialised tools or equipment required.
  - Estimate of total downtime.
  - If a Task List is known to exist for the job, it should be referenced to aid the maintenance planner.
- Proposed Priority for the work (i.e. how soon you would like the work completed), as per the below codes provided for reference. Note that a reasonable assessment of the true priority is crucial, since it will determine how soon the work is completed, and potentially cause disruption to planned work schedules.

Priority	Description	Work to be completed in
1	Immediate	0-1 days
2	High	1-7 days
3	Medium	8-28 days
4	Low	29-90 days
5	Shutdown	Next Shutdown

- Cause and Damage codes. These provide important information about common failure modes and will form the basis of future defect elimination work. Note that these codes will be reviewed (and updated where needed) later by the tradesperson after they complete the work.

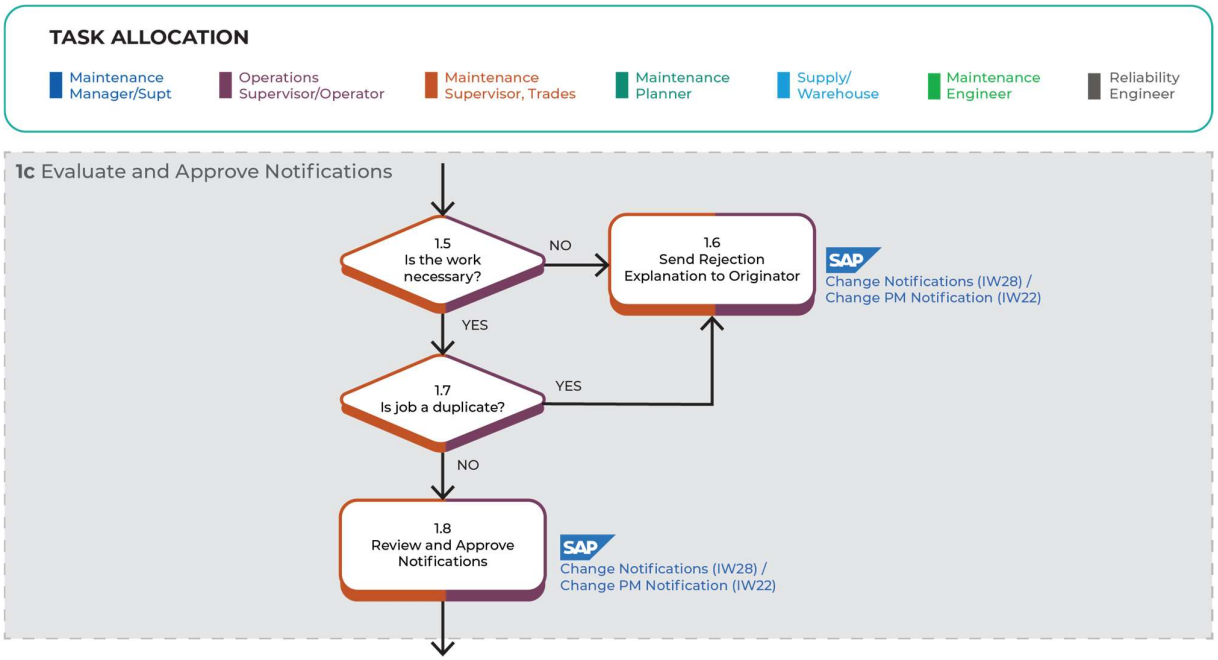
As a general rule, the more information included in the Notification, the better. See below example for of a good notification and refer to Appendix A in **SAP PM QRG – Create PM Notification** for further guidance.



Also consider attaching a photo or a sketch of the damage, if it will add value by explaining the situation more clearly. Refer to **SAP PM QRG – GOS Attachment to Notification** for details of how to do this.

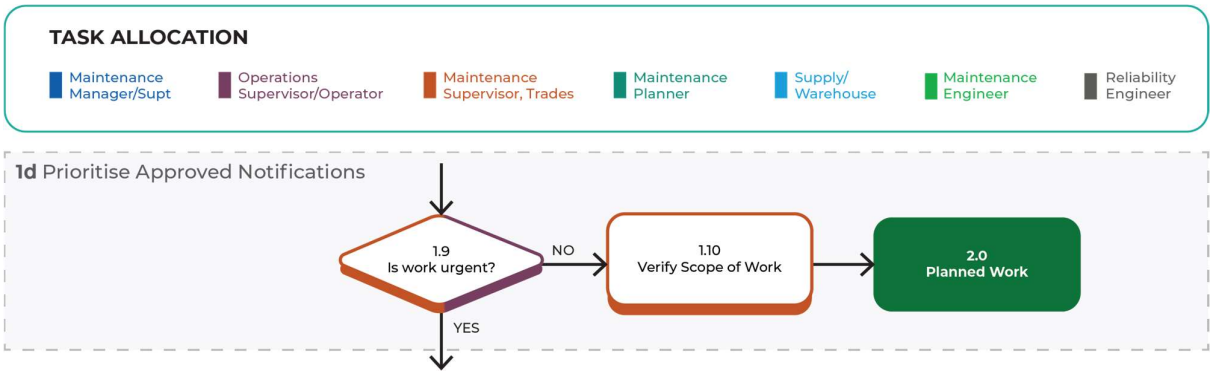
When?	After required work is identified.

1c Evaluate and Approve Notifications



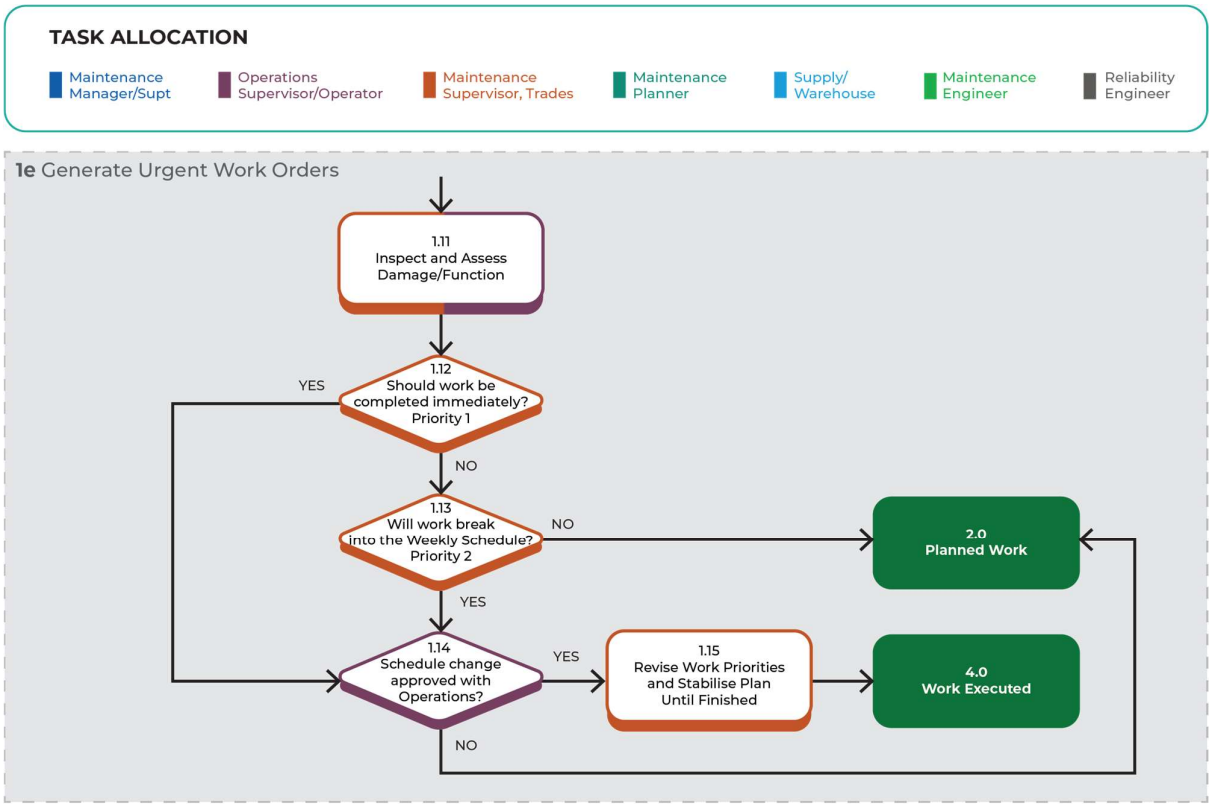
What?	Evaluate and approve/reject new Notifications.
Why?	Ensure that only necessary work is performed. Approve the Notification so that it can be converted into a Work Order, which allows planning to commence. It is critical that the right priority is assigned to the Notification as this could impact the current schedule.
Who?	• Maintenance Supervisors • Operations Supervisors
How?	Generate the list of outstanding Notifications in SAP using **Change Notifications (IW28)** or **Change PM Notification (IW22)**. The Supervisor shall approve the Notification if the work is necessary, is not a duplicate (i.e. another notification exists covering the work scope) and if it has enough information for the work to be completed, planned or scoped. The Supervisor should reject any Notifications that do not comply with the above requirements or has errors such as being raised against the wrong functional location or workgroup. • When rejecting Notifications always include a brief justification in the long text (the person who raised the notification will receive this in SAP, and it is important for them to know why it was rejected). Where important information or action is required, contact the originator or their Supervisor directly. • Where duplicates exist, capture the number of the duplicate Notification on the rejection note. Refer to **SAP PM QRG – Notification Put in Process & Reject** for detailed instruction on how to approve or reject notifications.
When?	Daily, at the beginning of each shift (before the **Daily Supervisors Review**).

1d Prioritise Approved Notifications



What?	Verify work is urgent, and if not send to the planning cycle (via assigned Priority).
Why?	By sending non-urgent work to the planning cycle, Supervisors are free to focus on their daily work schedule and any new urgent work.
Who?	• Maintenance Supervisor (consultation with Operations Supervisor, escalation to Superintendent as necessary). • Leading Hands/Tech Officers/Tradespersons to verify scope of works as assigned by Supervisor.
How?	Maintenance supervisors shall review the urgency of Notifications in SAP using **Change Notifications (IW28)** or **Change PM Notification (IW22)** and adjust priorities as needed, with consideration of impact to current Schedule and Operations: • Priority 1 - **Urgent** work that requires immediate attention, to be executed within 24hrs. • Priority 2 - **Urgent** work that should be executed in 1-7 day timeframe. • Priority 3, 4 & 5 **Non-Urgent** works enter the **2.0 Non-Urgent Work Planned** process (Planner workflow) to be planned and scheduled. Where job scoping is required for Non-Urgent work, the Maintenance Supervisor is responsible for completion and shall assign to a Leading Hand, Tech Officer or Tradesperson to perform job scoping activities to enable the planning process to commence thereafter. Completed job scoping information shall be captured in the Notification.
When?	Daily, prior to the **Daily Supervisors Review**.

1e Generate Urgent Work Orders



What?	Confirm that new urgent work is identified correctly and assess and manage any adverse impacts to the current maintenance schedule.
Why?	Completing Priority 1 and 2 urgent work in a timely manner is critical to complying with HSEC and obligatory requirements whilst achieving production targets. However, it is also important to understand that completing this new work can also have a negative impact on the agreed maintenance schedule (i.e. planned work not getting done). Therefore, the impacts to the maintenance schedule must be assessed and managed.
Who?	• Maintenance Supervisors (consultation with Operations Supervisors, escalation to Superintendents as necessary) • Maintenance Planner
How?	The Operations Supervisor shall confirm if work needs to be completed immediately as Priority 1 Urgent work (e.g. a breakdown on critical equipment impacting Operations) in consultation with the Maintenance Supervisor. When the work is not Priority 1, it becomes Priority 2 and will need to be completed in 1-7 days. The Maintenance Supervisor shall assess the impact to their weekly maintenance schedule, and where necessary identify exactly which scheduled work must be deferred.

	The Maintenance Supervisor must then assess the consequence of deferring the Scheduled work and determine whether the risk is acceptable. When assessing the risk of deferring work, the following should be considered: • Is it a statutory or compliance inspection? • Will it cause a safety risk? • Could it result in further degradation/damage, or risk of catastrophic failure? • Could it result in a breakdown that could impact production? • Can additional labour (external services) be sourced or overtime utilised to complete all works? • Will there be an opportunity to complete the work in the following week? If the risk of deferring scheduled work is not acceptable, the Supervisor recommends that the Priority 2 work be reassessed to a Priority 3, 4 or 5. The Operations Supervisor reviews this recommendation and approves the final Priority status (through agreement). The Maintenance Supervisor then reassigns (where appropriate) Priority 2 notifications as Priority 3, 4 or 5 in SAP using **Change Notifications (IW28)** or **Change PM Notification (IW22)** and it enters the **2.0 Non-urgent Work Planned** process (Planner workflow). Priority 1 and 2 (urgent) work does not enter the planner workflow and is managed by the maintenance team and executed under the **4.0 Work Executed** process.
When?	Daily, discussed in the **Daily Supervisors Review** (and processed after), or whenever Priority 1 or 2 urgent work is identified during a shift.

Daily Supervisors Review

Daily Supervisors Review	
Purpose	To review new notifications, agree on their Priorities, assess any impact or changes to current work schedules, and agree on the actions for the day. Review performance from the previous shifts.
Timing & Location	Daily at the beginning of the shift for 10 minutes, before Team Pre-Start Meetings. Held in the Maintenance Supervisors office.
Attendees	• Maintenance Supervisor (meeting lead) • Operations Supervisor
Preparations	All preparations by Maintenance Supervisor: • List of new notifications raised and filtered to ensure priorities are reasonable, duplicates and notifications with inadequate information rejected. • Maintenance Work Schedule for the current day and week. Review work completed on previous shift and scheduled work that was not completed. • Any shift reports or shift logs (operational or maintenance).
Agenda	1) Review any scheduled work not completed from the previous day (or shift) and assess impact on both Maintenance and Operations. 2) Review new Notifications since previous shift (focusing on Priority 1 and 2). Agree that priorities are correct. 3) Review shift logs to see if any other urgent work is mentioned/identified. 4) Review the day's work schedule and assess whether new Priority 1 jobs can be completed without impacting the scheduled work. Where the schedule will be impacted, identify which scheduled job(s) will be deferred. Agree on the updated daily work list. 5) Review the work schedule for the current week to assess whether any new Priority 2 jobs (or non-complete scheduled jobs from previous shift) can be completed without breaking the schedule. Where the schedule will be broken, identify which scheduled jobs will be deferred, and approve that the risk of this is acceptable. 6) Review the updated daily work list, and confirm that all necessary preparations (cleaning, isolations etc) are in hand. 7) Understand any impacts on Operations and agree mitigation plans as appropriate.
Meeting Outcomes	Confirmed list of work orders for the day for Team Pre-start Meeting (communicate any impacts to the current schedule caused by new Priority 1 or 2 work). Confirmed list of any scheduled work that has been deferred to be provided to the Maintenance Planner for re-scheduling. Any requirements from or impact on Operations identified, agreed and mitigated (as appropriate).

Measures of Success

The following are measures of success for sub-process **1. Work Identified, Registered & Prioritised**, as described in the Glencore Copper Asset Management Manual for **Maintenance Work Management** (Element 08).

Outstanding Notifications > 2 Days	
KPI Number	KPIGPM001
Definition	Notifications need to be approved or rejected within 48 hours from being created.
Performance Indicator	Zero outstanding Notifications older than 2 Days.
Calculation	Today's Date – Notification creation date
Rationale	This is important because it supports the empowerment of our employees for early identification of failures and effective communication. A low number of Outstanding Notifications indicates a good level of control, whereas one Outstanding Notification on a critical control can expose the organisation to material risk.

% of Proactive Notifications Raised	
KPI Number	KPIGM002
Definition	Proportion of priority 3, 4 & 5 Notifications from Operations & Maintenance versus the total number of Notifications raised in the same period.
Performance Indicator	>80%
Calculation	$\frac{\text{Total Priority 3, 4 \& 5 Notifications}}{\text{Total Number of Notifications}} \times 100$
Rationale	This metric drives positive behaviour from operators and trades to identify the priority of work as part of the initial assessment of work to be completed. A high number of priority 3, 4 and 5 notifications indicates an operation that is proactively identifying work that needs to be completed prior to it becoming an issue. A result less than 80% indicates an operation with a higher proportion of reactive maintenance.



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